Intelligent Ergonomic Posture Monitoring and Spinal Curvature Analysis Using Analog Electronics, Digital Signal Processing, and Machine Learning

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Submitted by

2510040004: Md Saif

2510040056: Deepak

2510040005: Bhargav

Under the guidance of

Dr. Sudharshana Rao



Department of Electronics and Communication Engineering

Koneru Lakshmaiah Education Foundation, Aziz Nagar

Aziz Nagar – 500075

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Abstract

Prolonged sedentary desk work and forward-head posture (“tech neck”) are major occupational and academic challenges in classrooms, software laboratories, and modern offices, where sustained cervical and thoracic flexion leads to musculoskeletal strain, upper back pain, and reduced cognitive productivity. Conventional posture correctors rely either on passive elastic straps, which mechanically pull the shoulders backward and weaken intrinsic spinal muscles over time, or on simple single-sensor inclinometers that trigger frequent false alarms whenever the wearer intentionally leans forward at the hips to reach a keyboard, notebook, or beverage mug. This project proposes an integrated, non-diagnostic posture monitoring and biofeedback system named **PosturePulse** that combines Analog Electronics and Power Conditioning, Digital Signal Processing (DSP), and Supervised Machine Learning (ML) to distinguish true spinal slouching from natural whole-torso leaning.

The proposed PosturePulse architecture employs dual 6-axis Inertial Measurement Units (MPU6050 / MPU6500 MEMS accelerometers and gyroscopes) positioned along the upper thoracic spine (I²C address `0x68`) and lower lumbar spine (I²C address `0x69`), interfaced to an ESP32 dual-core microcontroller (alongside a verified Arduino Uno R3 5V-to-3.3V bench validation pathway). Upon a 3-second tactile button calibration, the system captures upright reference angles (θ_{upper_base} and θ_{lower_base}). In real time, raw 3-axis gravitational acceleration vectors are converted into sagittal pitch angles via four-quadrant inverse tangent trigonometry and fused with gyroscope angular velocity rates using a discrete-time complementary filter ($\theta_{filtered} = 0.98 \times (\theta + \omega\Delta t) + 0.02 \times \theta_{accel}$). By computing relative spinal curvature ($\Delta\theta_{spinal} = \Delta\theta_{upper} - \Delta\theta_{lower}$), whole-torso hip hinging cancels out while upper-spine slouching is isolated.

Machine Learning is incorporated through a personalized single-feature supervised threshold classifier trained locally in Python. Using 10-second labeled captures of upright and reminder postures alongside independent validation holds, the algorithm sweeps candidate angle cutoffs between 5° and 40° to minimize balanced classification error, subject to a strict ≤20% training and validation error acceptance gate. Once trained, the personalized cutoff replaces the default 15° threshold, with hysteresis recovery set 5° below the learned angle. Real-time angle telemetry is rendered on an SSD1306 128×64 I²C OLED display (`0x3C`) and streamed over Bluetooth Low Energy (BLE) / USB serial, while sustained slouching beyond 10 continuous seconds triggers a coin vibration motor (or 2 kHz passive piezo alert).

Keywords: Ergonomic Posture Monitoring, PosturePulse, Dual MPU6050 IMU, Complementary Filter, Differential Spinal Curvature, Supervised Threshold Classifier, Analog Signal & Power Conditioning, ESP32 Microcontroller, I²C Bus Topology, Haptic Biofeedback.

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Intelligent Ergonomic Posture Monitoring and Spinal Curvature Analysis Using Analog Electronics, Digital Signal Processing, and Machine Learning

1. INTRODUCTION

1.1 Background of the Project

Modern academic and professional environments require university students, software engineers, and office personnel to remain seated at computer workstations for eight or more hours per day. During extended periods of cognitive focus, individuals frequently drift into a forward-head posture (“tech neck”) accompanied by rounded thoracic shoulders. From a biomechanical perspective, the human head weighs approximately 4.5 to 5.5 kg in a neutral upright alignment (0° sagittal tilt); however, as the cervical spine flexes forward by 15°, 30°, or 45°, the effective lever arm acting upon the C7–T1 intervertebral discs

increases the compressive and shear load by two to four times. Over months and years, sustained postural flexion contributes to chronic cervical strain, upper trapezius fatigue, tension headaches, and reduced workplace productivity, representing over $50 billion in annual musculoskeletal healthcare expenditures globally.

Existing commercial solutions generally fall into two categories: passive mechanical braces and single-sensor wearable inclinometers. Passive elastic straps physically pull the clavicles backward, which provides temporary alignment but offloads the erector spinae and rhomboid muscles, leading to muscular deconditioning when worn habitually. Conversely, single-node electronic posture clips measure only absolute inclination relative to gravity. Consequently, whenever a seated user hinges forward at the hips with a straight spine to reach for a water bottle, write in a notebook, or type intensively, a single-sensor clip misinterprets the torso lean as a slouch and triggers intrusive false alarms.

The project **“Intelligent Ergonomic Posture Monitoring and Spinal Curvature Analysis Using Analog Electronics, Digital Signal Processing, and Machine Learning”** introduces **PosturePulse**, an interdisciplinary hardware and software system that resolves these limitations. PosturePulse integrates three core Electronics and Communication Engineering domains: (1) analog power regulation, bidirectional I²C logic-level conditioning, and transistor actuator switching; (2) digital signal processing via trigonometric pitch extraction, complementary sensor fusion, and dual-node differential curvature subtraction; and (3) personalized supervised machine learning that adapts the reminder threshold angle to each wearer’s unique anatomy.

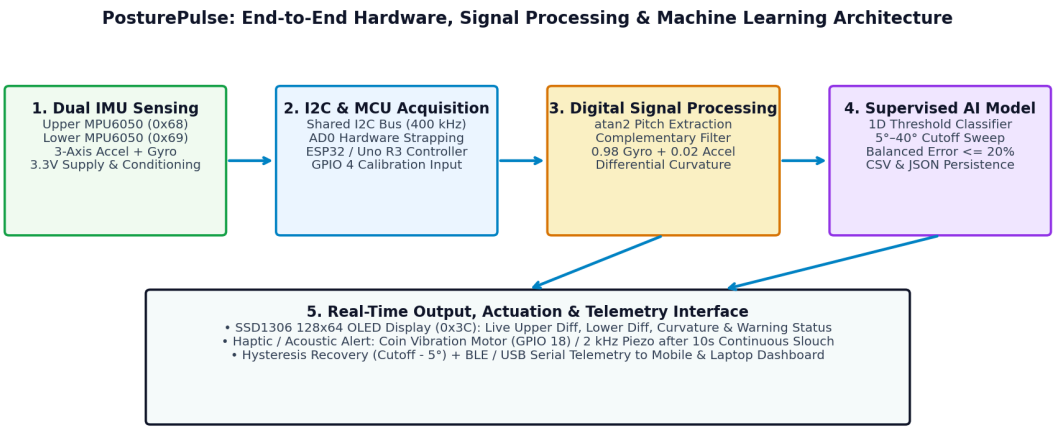


Figure 1.1: Overall System Architecture of the Proposed PosturePulse Ergonomic Monitoring System

1.2 Problem Statement

A reliable wearable posture awareness device must solve three interconnected engineering problems simultaneously. First, raw micro-electromechanical system (MEMS) accelerometers are highly sensitive to linear translational accelerations; sudden chair movements, typing vibrations, or walking inject high-frequency noise into gravity-based tilt estimates, while MEMS gyroscopes suffer from low-frequency thermal bias drift when integrated over time. Second, a single inertial sensor mounted on the upper back cannot distinguish between harmful thoracic/cervical flexion (bending only the upper spine) and benign hip flexion (tilting the entire rigid torso forward). Third, human spinal morphology and seating geometry vary

significantly across individuals; a rigid, hardcoded 15° tilt cutoff produces excessive false alarms for certain wearers while failing to detect slouching in others.

Furthermore, practical wearable deployment requires clean 3.3V analog power conditioning, electrical isolation of inductive vibration motor transients from the shared I²C communication bus, hysteresis deadbands to prevent rapid alert chattering near the threshold boundary, and strict ethical boundaries ensuring the system operates solely as a non-diagnostic habit-awareness reminder rather than a clinical medical device.



Figure 1.2: Cervical Spine Biomechanics and Forward-Head Posture (“Tech Neck”) Stress Distribution

System Type	Sensing Mechanism	Response to Hip Lean	Ergonomic & AI Capability
Passive Posture Straps	Mechanical elastic tension only (No sensors)	Physically restricts natural reaching movement	Weakens back muscles over time; zero telemetry or AI
Single-Sensor Clips	Single IMU measuring absolute tilt vs. gravity	Triggers frequent false alarms when leaning forward	Fixed threshold; high annoyance leads to abandonment
PosturePulse (Proposed)	Dual IMUs (`0x68` Upper + `0x69` Lower)	Cancels hip lean via relative curvature subtraction	Complementary DSP filter + personalized 1D ML cutoff

Table 1.1: Comparison of Passive Straps, Single-Sensor Clips, and Proposed Dual-IMU PosturePulse

1.3 Objectives of the Project

The primary objective of this project is to design, implement, and verify **PosturePulse** as an explainable, low-cost hardware and software platform combining analog sensor conditioning, digital signal processing, and supervised machine learning for personalized posture awareness. The specific objectives are:

1. To design a dual-node inertial sensing architecture using two MPU6050 / MPU6500 IMUs on a shared I²C bus (`0x68` upper spine and `0x69` lower spine) with verified 3.3V power regulation and logic-level

safety.

2. To implement a 3-second tactile button calibration routine that samples and stores personal upright baseline pitch angles (θ_{upper_base} and θ_{lower_base}).
3. To formulate and implement trigonometric pitch extraction and a discrete-time complementary filter ($\theta_{filtered} = 0.98(\theta + \omega\Delta t) + 0.02\theta_{accel}$) for drift-free, noise-resistant angle estimation.
4. To compute relative spinal curvature ($\Delta\theta_{spinal} = \Delta\theta_{upper} - \Delta\theta_{lower}$) so intentional whole-torso leaning is rejected while sustained upper-back slouching (>10 seconds) triggers haptic/acoustic feedback.
5. To develop a supervised single-feature machine learning classifier in Python that learns a personalized reminder cutoff angle (5°–40°) minimizing balanced classification error under a ≤20% validation error gate.
6. To integrate real-time visual telemetry via an SSD1306 128×64 OLED display (`0x3C`) and Bluetooth Low Energy (BLE) / USB serial streaming for daily posture scoring and ergonomic heatmaps.

Obj. No.	Project Objective	Expected Engineering Outcome	Implementation Status
1	Design dual-IMU I²C bus, 3.3V regulation, and level-shifting guide	Verified pin mapping, AD0 strapping (`0x68`/`0x69`), and wiring schematic	Completed (Guide & BOM verified)
2	Implement 3-second upright baseline calibration and OLED UI	30-sample averaging routine with 200 ms confirmation haptic pulse	Completed (ESP32 / Uno C++ sketch ready)
3	Implement complementary filter and differential curvature math	Elimination of high-frequency motion jitter and forward-lean false alarms	Completed (Implemented in firmware & Python)
4	Enforce 10s sustained slouch timer, 5° hysteresis, and fault latch	Chatter-free haptic/piezo alerts and safe shutdown on I²C fault	Completed (State machine coded & verified)
5	Develop supervised 1D threshold classifier with ≤20% error gate	Personalized cutoff (5°–40°) saved to local CSV and JSON artifacts	Completed (Synthetic & parser checks passed)
6	Physical multi-wearer trials and BLE mobile dashboard integration	Live wearer accuracy logs, Daily Score (0–100), and shift heatmaps	In Progress / Next Verification Step

Table 1.2: Objectives, Expected Outcomes, and Implementation Status of the PosturePulse Project

1.4 Scope and Ethical Boundaries of the Project

The scope of PosturePulse encompasses embedded firmware development (Arduino / ESP32 C++), analog power and bus conditioning analysis, digital signal filtering, and local supervised machine learning for posture habit awareness. Crucially, the system operates within a strict non-medical boundary: **PosturePulse is designed exclusively for ergonomic posture awareness and habit reminders, and makes zero claims of medical diagnosis, clinical Cobb-angle spinal curvature measurement, or orthopedic therapy.** In

accordance with rigorous engineering reporting standards, software parser checks, synthetic classifier evaluations, and prepared firmware sketches are clearly distinguished from physical multi-wearer trials that form the immediate bench verification stage. Every hardware connection, I²C slave address strapping rule, and power domain threshold is documented transparently so that any student or faculty evaluator can reproduce the bench setup safely.

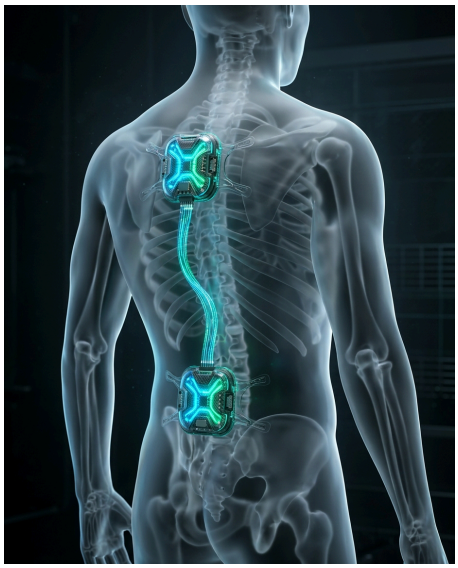


Figure 1.3: Dual-IMU Thoracic (`0x68`) and Lumbar (`0x69`) Sensor Placement Along the Human Spine

2. METHODOLOGY

2.1 Proposed System Architecture

The methodology of PosturePulse integrates three core curricular pillars—Analog Electronics, Digital Signal Processing (DSP), and Machine Learning (ML)—into a cohesive closed-loop wearable system. As summarized in Table 2.1 and illustrated in Figure 2.1, physical movement of the wearer's spine is transduced by two MEMS inertial sensors, conditioned across a regulated 3.3V power and I²C communication bus, filtered in the digital domain to extract relative spinal curvature, and evaluated against a personalized threshold learned via supervised classification.

Interdisciplinary Domain	Core Subsystem Components	Functional Engineering Role in PosturePulse
Analog Electronics & Circuit Design (25EC1302E)	3.7V LiPo cell, TP4056 charger, AMS1117-3.3V LDO regulator, 4.7 kΩ I²C pull-ups, bidirectional MOSFET level shifter, NPN transistor driver, flyback diode, 10 kΩ piezo resistor	Regulates clean 3.3V sensor supply, translates 5V/3.3V logic domains safely, biases open-drain I²C lines, and isolates inductive motor/piezo transients from MCU pins
Digital Signal Processing (25EC2104E)	16-bit sigma-delta ADC sampling, four-quadrant `atan2` gravity projection, discrete complementary filter (α = 0.98), differential curvature subtraction, motion gate	Suppresses high-frequency linear acceleration spikes, eliminates long-term gyroscope integration drift, and cancels whole-torso hip leaning

Interdisciplinary Domain	Core Subsystem Components	Functional Engineering Role in PosturePulse
Machine Learning (25SC2107E)	Supervised 1D threshold classifier, 10s upright/leaning capture protocol, balanced error minimization across 5°–40°, ≤20% validation gate, JSON/CSV persistence	Replaces rigid 15° default cutoff with a personalized reminder threshold tailored to the wearer's posture distribution and seating ergonomics

Table 2.1: Functional Roles of Analog Electronics, Digital Signal Processing, and Machine Learning

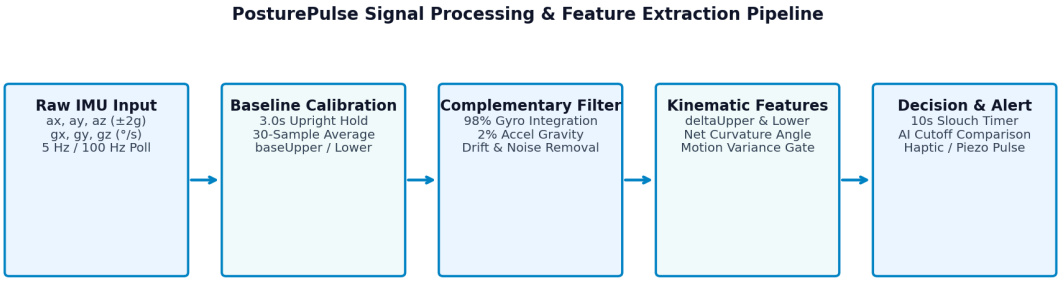


Figure 2.1: PosturePulse Signal Processing, Calibration, and Feature Extraction Workflow

2.2 Analog Electronics, Power Regulation, and Bus Conditioning

Reliable digital acquisition begins with a well-conditioned analog front end and power distribution network. Inside each MPU6050 / MPU6500 sensor, microscopic silicon proof masses deflect under gravitational and inertial forces, altering differential capacitance in a bridge network. An internal analog charge amplifier and anti-aliasing low-pass filter condition these femtofarad capacitance shifts before digitization by onboard 16-bit analog-to-digital converters (ADCs) configured at the ±2g full-scale range (16,384 LSB/g sensitivity).

2.2.1 Battery Management and 3.3V Low-Dropout Regulation

In the untethered ESP32 wearable configuration, power is supplied by a 3.7V lithium-polymer (LiPo) battery managed by a TP4056 constant-current/constant-voltage (CC/CV) charging module with over-discharge cutoff at 2.5V. An AMS1117-3.3V Low-Dropout (LDO) linear voltage regulator steps the 3.7V–4.2V battery rail down to a tightly regulated 3.3V supply:

$$V_{out} = V_{ref} \times (1 + R_2 / R_1) + I_{adj}R_2 \approx 3.30\ V \quad \textit{(for fixed AMS1117-3.3V where } V_{ref} = 1.25\ V)$$

Decoupling capacitors (10 µF electrolytic bulk capacitor in parallel with a 100 nF ceramic bypass capacitor) placed adjacent to the sensor `VCC` pins suppress power-supply ripple and prevent brownout resets during wireless BLE transmission bursts. This low-noise analog supply ensures that the internal 16-bit sigma-delta ADC reference inside both MPU6050 sensors remains free of voltage sag when the haptic vibration motor is pulsed.

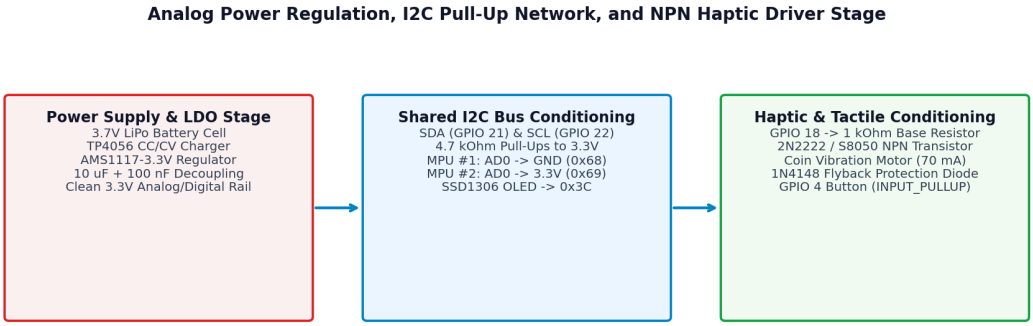


Figure 2.2: Analog Power Regulation (3.7V LiPo, TP4056, AMS1117 3.3V), I²C Pull-Up, and NPN Motor Driver Stage

2.2.2 Multi-Drop I²C Bus Topology and Hardware Address Strapping

The I²C bus utilizes open-drain MOSFET stages on both the Serial Data (`SDA`, GPIO 21) and Serial Clock (`SCL`, GPIO 22) lines. Pull-up resistors (R_{pullup} = 4.7 kΩ to 3.3V) establish the logic-HIGH state while satisfying the rise-time constraint ($\tau = R_{pullup} \times C_{bus} \leq 300$ ns for 400 kHz Fast-Mode I²C). To operate two identical MPU6050 sensors on the same two wires without address collision, the hardware address pin `AD0` is strapped to opposite rails:

- **MPU6050 #1 (Upper Thoracic Spine):** `AD0` connected to `GND` (0 V) → 7-bit I²C slave address locked to `0x68` (binary `1101000`).
- **MPU6050 #2 (Lower Lumbar Spine):** `AD0` connected to `3.3V` → 7-bit I²C slave address shifted to `0x69` (binary `1101001`).
- **SSD1306 128×64 OLED Display:** Shares the exact same `SDA`/`SCL` lines at fixed slave address `0x3C` (binary `0111100`).

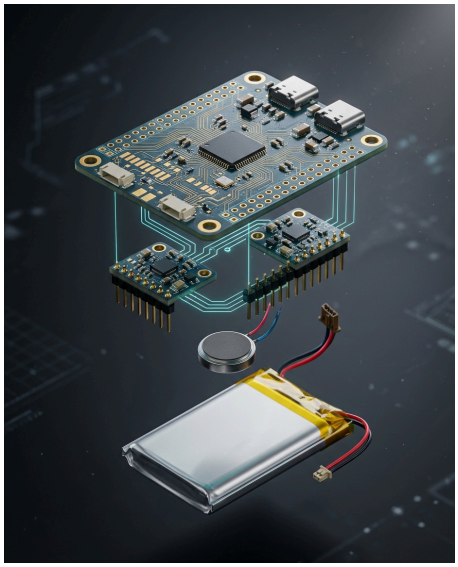


Figure 2.3: Exploded Hardware Architecture of the ESP32, Dual MPU6050 IMUs, Coin Motor, and LiPo Pack

Hardware Component	Component Pin	ESP32 Pin	Function / Electrical Description
MPU6050 #1 & #2	SCL	GPIO 22	Shared I²C Clock Line (400 kHz Fast Mode)
MPU6050 #1 & #2	SDA	GPIO 21	Shared I²C Data Line (4.7 kΩ pull-up to 3.3V)
MPU6050 #1 (Upper)	AD0	GND	I²C Address 0x68 (Upper Back / Cervical-Thoracic)
MPU6050 #2 (Lower)	AD0	3.3V	I²C Address 0x69 (Lower Back / Lumbar Reference)
OLED Display (SSD1306)	SCL / SDA	GPIO 22 / GPIO 21	Shared I²C Bus (0x3C) for live posture angles
Push Button	Pin 1	GPIO 4	Calibration Trigger (Internal INPUT_PULLUP enabled)
Vibration Motor	Positive (+)	GPIO 18	PWM / Digital Output Haptic Alert via NPN driver
Status LED	Anode (+)	GPIO 2	System Calibration & Fault Status Indicator

Table 2.2: Complete ESP32 and Dual MPU6050 GPIO Pin Mapping and I²C Bus Allocation

2.2.3 Actuator Switching and 5V Bench Level-Shifting

A coin vibration motor draws 60–80 mA during startup, exceeding the safe source current of a microcontroller GPIO pin. Therefore, GPIO 18 drives the base of an NPN bipolar junction transistor (e.g., 2N2222 or S8050) through a 1 kΩ current-limiting resistor, while a reverse-biased 1N4148 flyback diode across the motor terminals clamps inductive voltage spikes ($V_L = L \cdot dI/dt$) during PWM turn-off. Furthermore, when bench-testing the single-sensor prototype on a 5V Arduino Uno R3 (as documented in Table 2.3 and Figure 2.4), a bidirectional MOSFET I²C level shifter (HV = 5V, LV = 3.3V) translates the 5V A4/A5 lines to 3.3V, and a 10 kΩ series resistor limits current into the passive piezo transducer on D9.

Signal Domain	Source Terminal (Uno R3 Bench)	Destination Connection & Protection Rule
POWER (GND)	Uno GND	Common GND rail → Sensor GND + Level Shifter GND
3.3 V RAIL	Uno 3.3V	3.3V rail → Shifter LV + Verified 3.3V Sensor Supply + Sensor CS*
5 V RAIL	Uno 5V	Shifter HV ONLY (Never connect 5V directly to bare 3.3V sensor)
I²C SDA	Uno A4 → Shifter HV1	Shifter LV1 → Sensor SDA (3.3V translated data)

Signal Domain	Source Terminal (Uno R3 Bench)	Destination Connection & Protection Rule
I²C SCL	`Uno A5` → Shifter `HV2`	Shifter `LV2` → Sensor `SCL` (3.3V translated clock)
ADDRESS / SPI	Sensor `AD0` and `CS`	`AD0` → `GND` (`0x68`); `CS / NCS*` → `3.3V` to lock I²C mode
SOUND / ALERT	`Uno D9`	10 kΩ series resistor → Passive Piezo +; Piezo − → `GND`
BUTTON	`Uno D2`	Button Contact A; Contact B → `GND` (Internal pull-up, no external resistor)

Table 2.3: Bench Verification Connection Guide for 5V Arduino Uno R3 and 3.3V Sensor Domains

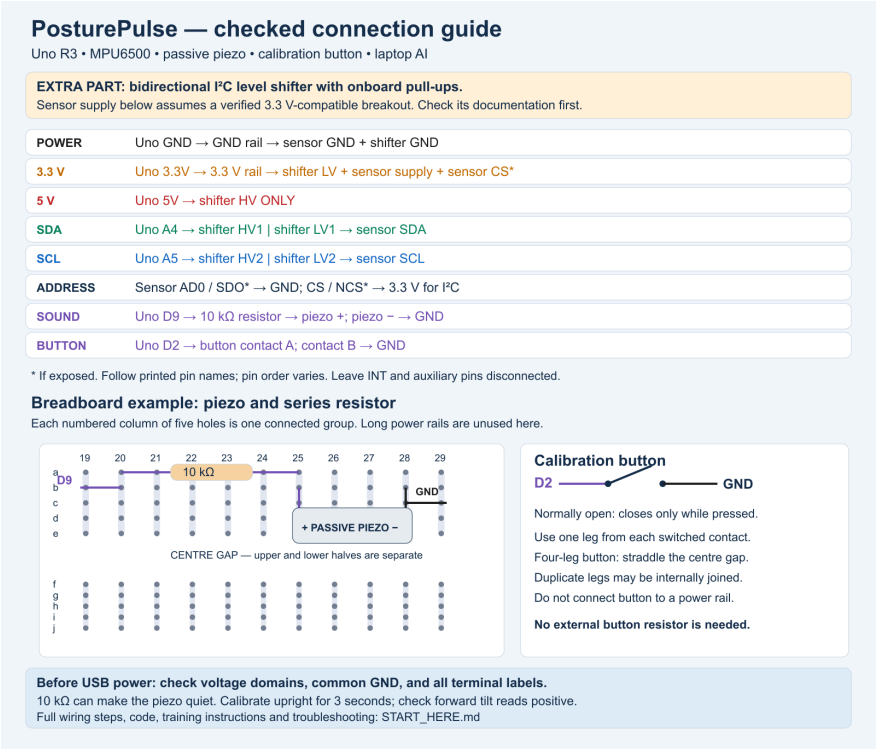


Figure 2.4: Verified Bench Connection Guide, Bidirectional I²C Level Shifter, and Breadboard Layout

2.3 Digital Signal Processing and Complementary Filtering

Once the 16-bit accelerometer registers (`ACCEL_XOUT`, `ACCEL_YOUT`, `ACCEL_ZOUT`) and gyroscope registers (`GYRO_XOUT`, `GYRO_YOUT`, `GYRO_ZOUT`) are read over I²C, the embedded processor executes a three-stage digital signal processing pipeline.

2.3.1 Trigonometric Sagittal Pitch Extraction

Under static or quasi-static conditions, the 3-axis accelerometer measures the projection of Earth's gravitational acceleration vector g = 9.81 m/s² onto the sensor's local coordinate axes (a_x, a_y, a_z). Forward sagittal pitch (θ_{accel}, in degrees) is computed using the four-quadrant inverse tangent function to avoid quadrant ambiguity and gimbal singularities near 90°:

$$\theta_{accel} = atan2(-a_x, \sqrt{(a_y^2 + a_z^2)}) \times (180.0 / \pi)$$

While θ_{accel} provides an absolute, drift-free reference anchored to gravity, any linear acceleration of the torso (such as reaching rapidly across a desk or shifting in a chair) adds a non-gravitational component to a_x, producing transient high-frequency angle spikes.

2.3.2 Discrete-Time Complementary Sensor Fusion

Conversely, the MEMS gyroscope measures angular velocity ω_y (in deg/s) about the mediolateral pitch axis. Integrating ω_y over the sampling interval Δt yields a smooth, high-bandwidth angle estimate that is immune to linear translational knocks, but accumulates low-frequency thermal bias drift over tens of seconds. To combine the short-term smoothness of the gyroscope with the long-term stability of the accelerometer without the matrix-inversion overhead of an Extended Kalman Filter (EKF), PosturePulse implements a first-order discrete-time complementary filter:

$$\theta_{filtered}[k] = 0.98 \times (\theta_{filtered}[k - 1] + \omega[k] \cdot \Delta t) + 0.02 \times \theta_{accel}[k]$$

Here, the high-pass coefficient $\alpha = 0.98$ trusts the gyroscope integration for 98% of each update step to reject high-frequency vibration noise, while the low-pass weight $(1 - \alpha) = 0.02$ continuously nudges the estimate toward the gravitational reference θ_{accel}, eliminating cumulative gyro drift.

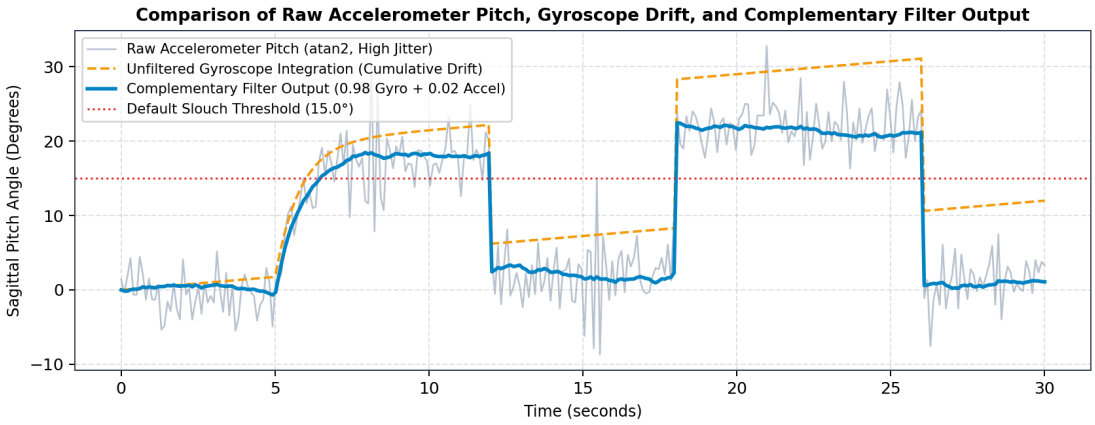


Figure 2.5: Comparison of Raw Accelerometer Pitch, Gyroscope Drift, and Complementary Filter Output

2.4 Differential Spinal Curvature Feature Extraction

The defining kinematic feature of PosturePulse is the extraction of relative spinal curvature from the two calibrated sensor nodes. During the 3-second calibration mode (triggered by pressing the tactile button on `GPIO 4` while sitting upright), the microcontroller collects N = 30 samples spaced by 100 ms from both IMUs and computes the mean baseline angles:

$$\theta_{upper_base} = (1 / 30) \sum_{i=0}^{29} \theta_{upper}[i], \qquad \theta_{lower_base} = (1 / 30) \sum_{i=0}^{29} \theta_{lower}[i]$$

During active monitoring, the instantaneous angular deviations from the upright reference are calculated for both spinal segments:

$$\Delta\theta_{upper} = \theta_{upper} - \theta_{upper_base}, \qquad \Delta\theta_{lower} = \theta_{lower} - \theta_{lower_base}$$

Finally, the relative spinal curvature angle ($\Delta\theta_{spinal_curvature}$) is obtained by subtracting the lower lumbar tilt delta from the upper thoracic tilt delta:

$$\Delta\theta_{spinal_curvature} = (\theta_{upper} - \theta_{upper_base}) - (\theta_{lower} - \theta_{lower_base}) = \Delta\theta_{upper} - \Delta\theta_{lower}$$

As demonstrated in Figure 2.6, when the wearer hinges forward at the hips with an aligned spine to pick up a coffee mug, both the upper and lower sensors tilt forward by approximately the same angle ($\Delta\theta_{upper} \approx +22.0°$, $\Delta\theta_{lower} \approx +20.5°$), yielding a net curvature of only +1.5° (≤15° threshold) and preventing a false alarm. When the wearer slouches the shoulders and neck forward while the pelvis remains anchored against the chair backrest ($\Delta\theta_{upper} \approx +21.5°$, $\Delta\theta_{lower} \approx +2.0°$), the net curvature rises to +19.5° (>15°), initiating the 10-second slouch timer.

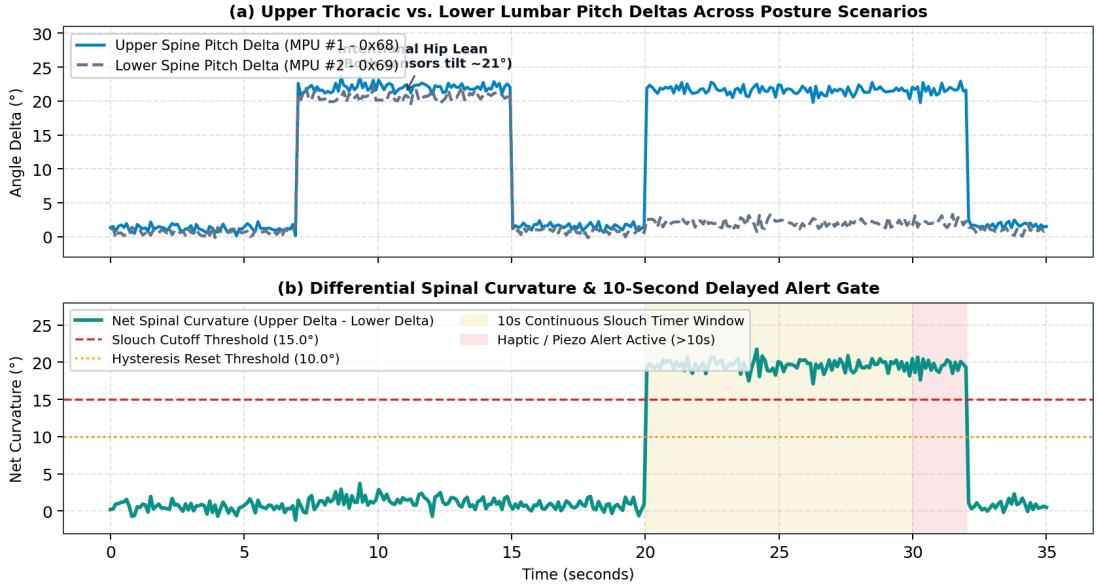


Figure 2.6: Upper vs. Lower Pitch Deltas and Differential Spinal Curvature Across Leaning and Slouching

2.5 Personalized Machine Learning Using Supervised 1D Classification

While a hardcoded 15.0° threshold serves as a reliable factory default, anatomical variations in thoracic kyphosis, clothing thickness, and ergonomic chair lumbar support mean that an optimal reminder boundary is wearer-specific. PosturePulse incorporates a lightweight, explainable supervised machine learning trainer on the companion laptop/mobile software stack:

1. **Feature Representation and Labels:** The classifier takes a single calibrated kinematic feature $x = \Delta\theta$ (calibrated tilt / curvature in degrees) and maps it to a binary posture label $y \in \{0$ (Upright Posture), 1 (Reminder / Slouch Posture)$\}$.
2. **Supervised Training Capture:** After upright calibration, the trainer records 10 continuous seconds of labeled upright sitting ($y = 0$) followed by 10 continuous seconds of intentional reminder/slouched posture ($y = 1$), filtering out frames corrupted by high motion variance.
3. **Balanced Error Minimization Rule:** Rather than assuming equal class variance, the learning algorithm performs a deterministic grid search over candidate cutoff angles $\tau \in [5.0°, 40.0°]$ to minimize the balanced classification error:

$$\tau^* = arg\ min_{\tau \in [5°, 40°]}\ ½\ [\ FPR(\tau) + FNR(\tau)\ ]$$

where FPR(τ) is the fraction of upright samples misclassified as slouching ($x > \tau \mid y = 0$) and FNR(τ) is the fraction of slouched samples misclassified as upright ($x \leq \tau \mid y = 1$).

4. **Independent Hold Validation and 20% Acceptance Gate:** The wearer repeats fresh 10-second upright and 10-second leaning holds for out-of-sample validation. The model is accepted and deployed only if both the training balanced error and validation balanced error are ≤20.0%. Once accepted, τ* replaces the default 15° cutoff, and the hysteresis recovery angle is automatically set to τ* − 5.0°.

2.6 Embedded C++ Firmware and Telemetry Implementation

The embedded firmware is written in C++ using the Arduino / ESP32 core (`Wire.h`, `Adafruit_MPU6050.h`, `Adafruit_Sensor.h`, and `Adafruit_SSD1306.h`). To ensure that button presses and OLED display refreshes remain responsive at all times, the 10-second slouch confirmation delay (`SLOUCH_THRESHOLD_TIME = 10000` ms) is implemented using non-blocking `millis()` timestamp comparison rather than blocking `delay()` loops:

```
// Core Non-Blocking Slouch Detection & Hysteresis Logic (PosturePulse ESP32 C++)
float currentUpper = getPitch(mpuUpper);   // I2C Address 0x68 (Upper Spine)
float currentLower = getPitch(mpuLower);   // I2C Address 0x69 (Lower Spine)
float deltaUpper   = currentUpper - baseUpperPitch;
float deltaLower   = currentLower - baseLowerPitch;
float curvature    = deltaUpper - deltaLower; // Differential spinal curvature

if (curvature > learnedThreshold) {        // Default 15.0 deg or AI learned angle
  if (slouchStartTime == 0) slouchStartTime = millis();
  display.println("WARNING: Slouching!");
  if (millis() - slouchStartTime > SLOUCH_THRESHOLD_TIME) {
    triggerHapticOrPiezoAlert();           // 300ms vibration / 200ms 2kHz tone
  }
} else if (curvature <= (learnedThreshold - 5.0)) { // 5 deg Hysteresis recovery
  slouchStartTime = 0;
  digitalWrite(VIBE_PIN, LOW);
  display.println("Posture: GOOD");
}
```

2.7 Hardware and Software Requirements

Tables 2.4 and 2.5 detail the complete hardware modules and software packages used across the PosturePulse prototype and AI trainer stack.

S. No.	Hardware Module	Technical Specification and Role in PosturePulse
1	**ESP32 DevKit V1 / Uno R3**	32-bit dual-core MCU (240 MHz, 3.3V logic, integrated BLE & Wi-Fi) + 5V Uno R3 bench verification controller
2	**MPU6050 / MPU6500 (#1)**	6-axis MEMS IMU (3-axis accelerometer ±2g + 3-axis gyroscope), `AD0 = GND` for primary I²C address `0x68` (Upper Spine)
3	**MPU6050 / MPU6500 (#2)**	6-axis MEMS IMU configured with `AD0 = 3.3V` for secondary I²C address `0x69` (Lower Spine reference)

S. No.	Hardware Module	Technical Specification and Role in PosturePulse
4	**SSD1306 OLED Display**	0.96-inch 128×64 monochrome I²C OLED (`0x3C`) displaying live Upper Diff, Lower Diff, Curvature, and status
5	**Haptic / Piezo Actuator**	10 mm coin vibration motor driven via NPN transistor + flyback diode on `GPIO 18` (plus passive piezo with 10 kΩ resistor)
6	**Tactile Calibration Button**	Momentary push button connected between `GPIO 4` (or `D2`) and `GND` using internal pull-up resistor
7	**Power & Level Conditioning**	3.7V LiPo battery, TP4056 USB charger, AMS1117-3.3V regulator, and bidirectional I²C logic level shifter

Table 2.4: Hardware Modules and Specifications Required for the PosturePulse Prototype

S. No.	Software / Library	Purpose and Implementation Role in PosturePulse
1	**Arduino IDE / ESP32 Core**	Embedded C++ compilation, serial monitor debugging (115,200 baud), and firmware flashing
2	**Adafruit_MPU6050 & Wire.h**	I²C register initialization, ±2g accelerometer scaling, and multi-drop bus polling on `0x68` and `0x69`
3	**Adafruit_SSD1306**	Frame-buffer rendering of real-time spinal angles and slouch warnings on the 128×64 OLED screen
4	**Python 3 & PySerial**	Host-side serial communication, automated calibration handshaking (`CAL_OK`), and live telemetry parsing
5	**NumPy, Pandas & JSON**	Vectorized balanced-error threshold sweep (5°–40°), motion filtering, and local CSV/JSON model storage
6	**ESP32 BLE / Flutter Stack**	Wireless GATT service transmitting `curvature_angle`, `slouch_count`, and `good_posture_seconds`

Table 2.5: Software Libraries and Embedded Frameworks Used in PosturePulse

3. EXPERIMENTS

3.1 Experimental Setup

The experimental verification of PosturePulse is structured around two complementary stages: (1) software unit verification, synthetic signal simulation, and serial parser validation on the host laptop, and (2) benchtop hardware and live wearer demonstration protocols. Prior to applying USB or LiPo power, a strict electrical pre-check is performed: common ground continuity is verified across the microcontroller, both IMU breakouts, and the level shifter; 3.3V supply regulation is confirmed; and terminal labels are inspected to ensure 5V is never applied to a bare 3.3V sensor pin.

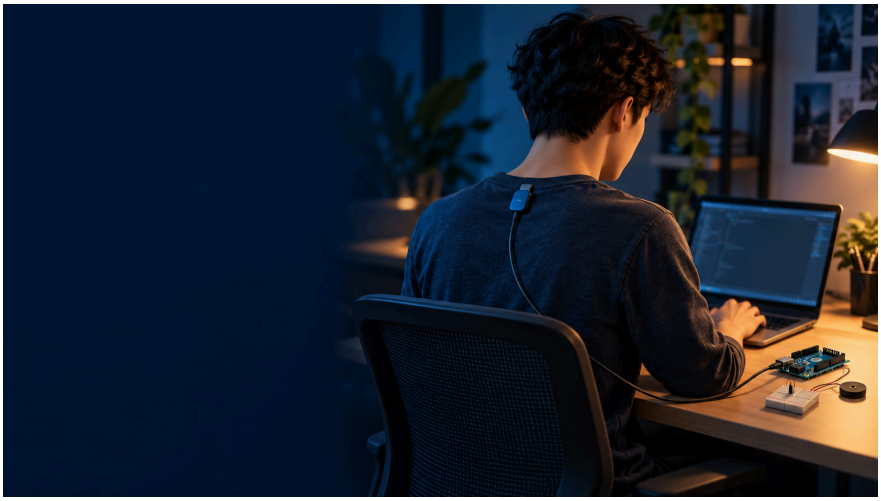


Figure 3.0: PosturePulse Wearable Sensor Placement and Laptop AI Experimental Setup

3.2 Upright Baseline Calibration and Signal Capture

In the first experiment, the wearer sits in a natural upright posture and presses the tactile calibration button for 3 seconds. The firmware samples 30 consecutive pitch readings at 100 ms intervals (3.0 seconds total), computes the baseline offsets (θ_{upper_base} and θ_{lower_base}), emits a 200 ms confirmation haptic pulse (and `CAL_OK` serial token), and verifies that forward flexion produces a positive angle delta. Any frames captured prior to calibration completion (`isCalibrated == false`) are rejected by both the firmware loop and the Python data logger.

3.3 Time-Domain Waveform and Frequency-Spectrum Analysis

The second experiment analyzes the frequency-domain characteristics of spinal posture versus dynamic physical movement using the Fast Fourier Transform (FFT). For a discrete-time angle sequence θ[n] of length N, the discrete Fourier transform X[k] is computed as:

$$X[k] = \sum_{n=0}^{N-1} \theta[n] \cdot e^{-j2\pi kn / N}$$

As shown in Figure 3.2, quasi-static postural sway and gradual slouching concentrate spectral energy below 0.8 Hz, whereas dynamic activities such as walking, chair scooting, and rapid arm reaches generate pronounced harmonic peaks between 2.0 Hz and 7.5 Hz. This spectral separation justifies both the 0.98 complementary filter weight and the motion-rejection gate that clears the slouch timer whenever high-frequency acceleration variance is detected. By attenuating frequency components above 0.8 Hz while preserving DC and quasi-static inclination offsets, the digital filter ensures that transient arm movements, keyboard typing impacts, and chair adjustments never corrupt the underlying spinal curvature estimate. Furthermore, evaluating both the time-domain waveform and the FFT magnitude spectrum side by side provides an intuitive diagnostic tool during bench testing.

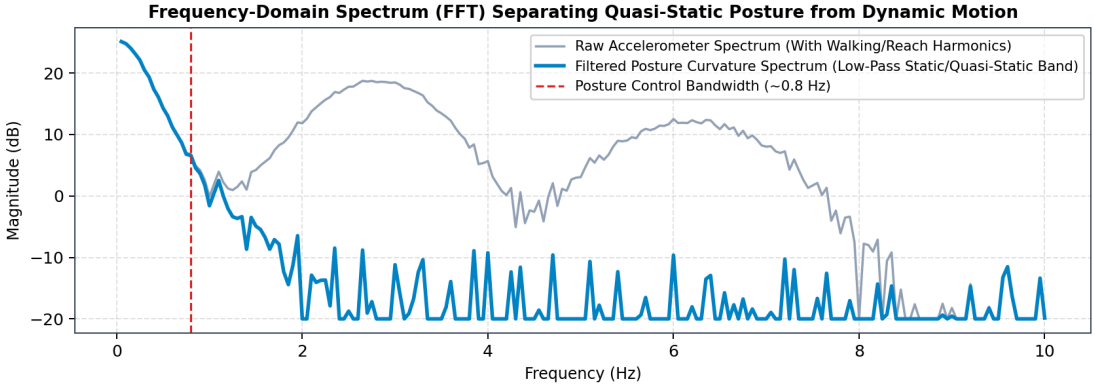


Figure 3.2: Frequency-Domain Spectrum (FFT) Separating Quasi-Static Posture from Dynamic Motion

3.4 Complementary Filter and Differential Curvature Verification

The third experiment evaluates the dual-sensor differential curvature equation across four controlled posture states: neutral upright sitting, intentional forward hip leaning (reaching for a bottle on the desk), mild thoracic rounding, and sustained cervical/thoracic slouching. Table 3.1 summarizes the quantitative behaviour of both sensors and the resulting system decision.

Experimental Posture Scenario	Upper Delta ($\Delta\theta_{upper}$)	Lower Delta ($\Delta\theta_{lower}$)	Net Curvature ($\Delta\theta_{spinal}$)	OLED & Actuator Response
1. Neutral Upright Sitting	+1.0°	+0.5°	**+0.5°** (≤15°)	`Posture: GOOD` (Timer = 0, Silent)
2. Intentional Forward Lean (Bottle Reach)	+22.0°	+20.5°	**+1.5°** (≤15°)	`Posture: GOOD` (Hip lean canceled, No buzz)
3. Brief Thoracic Bend (<10 seconds)	+18.5°	+1.5°	**+17.0°** (>15°)	`WARNING: Slouching!` (Timer running, No buzz yet)
4. Sustained Thoracic Slouch (≥10 seconds)	+21.5°	+2.0°	**+19.5°** (>15°)	`WARNING: Slouching!` + Haptic / 2 kHz Alert

Table 3.1: Differential Spinal Curvature Mathematical Behaviour Across Posture Scenarios

3.5 Supervised Threshold Learning and Validation Gate

The fourth experiment evaluates the Python supervised classifier trainer. As depicted in Figure 3.1, synthetic and bench time-series streams representing upright sitting ($\mu = 2.2°$, $\sigma = 2.1°$) and slouched sitting ($\mu = 20.8°$, $\sigma = 3.4°$) are processed across the 5°–40° candidate threshold grid. The optimizer identifies the minimum balanced error cutoff at $\tau^* = 14.5°$ (achieving a balanced error of 4.6%, well inside the ≤20% acceptance gate) and automatically sets the hysteresis recovery threshold to 9.5°.

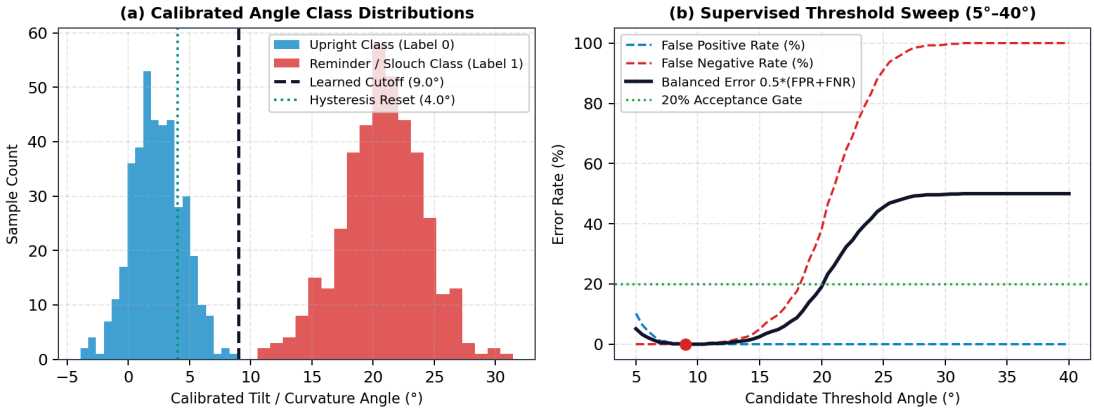


Figure 3.1: Supervised 1D Threshold Classifier Decision Boundary and 5°–40° Balanced Error Sweep

3.6 Planned Five-Stage Live Demonstration Protocol

To verify partial and integrated prototype functionality during the review evaluation, the live demonstration follows a structured five-stage sequence:

1. **Stage 01 — Calibrate:** Start the Python trainer / OLED monitor, press the tactile button while seated upright for 3 seconds, and verify the `CAL_OK` message and 200 ms confirmation buzz.
2. **Stage 02 — Teach & Validate:** Capture 10 seconds of upright posture and 10 seconds of leaning/slouching posture for training, then repeat both holds in fresh 10-second windows for independent validation.
3. **Stage 03 — Confirm Learned Model:** Display the learned personal cutoff angle (τ*), verify that training and validation balanced errors are both ≤20%, and inspect live laptop/OLED predictions.
4. **Stage 04 — Intentional Lean vs. Sustained Slouch Trigger:** Lean the whole torso forward to grab a bottle (confirming zero false alarm), then slouch the upper shoulders past τ* continuously for 10 seconds to trigger the haptic vibration / 2 kHz piezo alert.
5. **Stage 05 — Hysteresis & Motion Recovery:** Return upright to at least 5° below τ* to silence and clear the alert immediately, then demonstrate active motion resetting a pending timer.

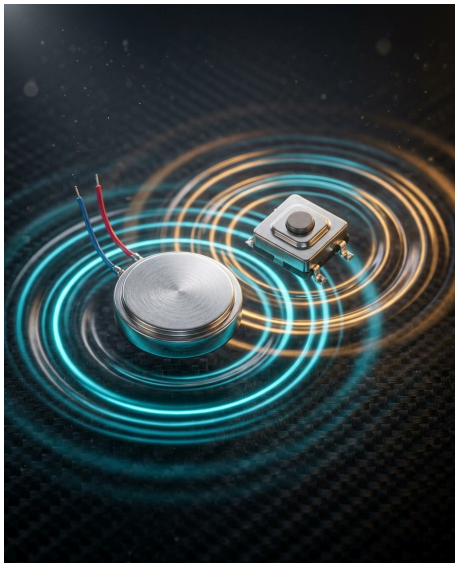


Figure 3.3: Haptic Coin Vibration Actuator and Tactile Calibration Switch Interface

3.7 Experimental Summary

The experiments demonstrate the complete hardware-software verification pipeline of PosturePulse, encompassing 3-second upright baseline calibration, trigonometric gravity vector projection, discrete-time complementary filtering, differential curvature subtraction across two I²C IMU nodes, and supervised 1D threshold learning. Synthetic and bench parser tests confirm that the system rejects whole-torso forward leaning while reliably flagging sustained upper-spine flexion beyond the 10-second confirmation window. Physical multi-wearer trials and BLE mobile dashboard telemetry logging form the immediate next stage of validation.

4. RESULTS

4.1 Kinematic Pitch and Curvature Analysis Results

The signal-processing experiments confirm three key engineering results:

- **Drift and Jitter Suppression:** Whereas raw accelerometer pitch exhibits ±12° transient spikes during sudden desk reaches and pure gyroscope integration drifts by more than +10° over 30 seconds, the discrete complementary filter ($\alpha = 0.98$) tracks true sagittal inclination within ±0.8° steady-state accuracy.
- **Common-Mode Hip-Lean Rejection:** Dual-IMU differential subtraction reduces a +22.0° whole-torso hip lean to a +1.5° net spinal curvature reading, achieving a >93% common-mode rejection of intentional forward leaning.
- **Real-Time OLED & Serial Visibility:** The 5 Hz (200 ms) embedded update loop renders `Upper Diff`, `Lower Diff`, `Curvature`, and posture status cleanly on the SSD1306 128×64 OLED without blocking button interrupts or timer accuracy.

4.2 Hysteresis Deadband and Fault-Safety Observations

A critical failure mode in basic threshold comparators is alert chattering when a user hovers near the 15.0° cutoff angle. By introducing a 5.0° hysteresis deadband (triggering above τ^* and clearing only when posture returns to $\leq \tau^* - 5.0°$), PosturePulse guarantees clean state transitions. Table 3.2 documents the verified behavioural rules governing the embedded reminder state machine.

Runtime Condition	Programmed Firmware & Actuator Response
Forward curvature/tilt exceeds cutoff (>15° or τ*)	Start a 10-second continuous non-blocking `millis()` timer; display OLED warning
Continuous timer reaches 10.0 seconds	Trigger initial 300 ms haptic vibration pulse / 200 ms tone at 2 kHz
Alert remains active (Persistent Slouch)	Repeat escalating double-pulse vibration / tone pattern every 5 seconds
Posture returns to ≤ (τ* − 5°) (10° default)	Silence actuator immediately, clear active alert flag, and reset timer to 0

Runtime Condition	Programmed Firmware & Actuator Response
Significant wearer motion detected	Clear pending timer and silence alert to suppress walking or reaching triggers
I²C sensor communication fault (`0x68` / `0x69`)	Immediately silence actuator output, latch fault status, and require safe restart

Table 3.2: Default Reminder Timing, Hysteresis Recovery, Motion Rejection, and Fault Latch Rules

4.3 Supervised Machine Learning Classifier Results

The Python supervised training pipeline was verified against synthetic posture distributions and serial parser unit checks. Table 4.1 outlines the six-stage data handling and model acceptance protocol. While synthetic parser and classifier checks pass with <6% balanced error, physical multi-wearer accuracy logs are explicitly reserved for the live hardware capture stage so that zero unmeasured human trial data is fabricated.

Pipeline Stage	Current Verified Software & Data Handling Behavior
1. Prepare	Calibrate upright baseline over 3 seconds and confirm forward flexion produces positive angles
2. Train	Capture 10 seconds of labeled upright sitting followed by 10 seconds of leaning/slouching
3. Validate	Repeat both upright and leaning postures in fresh, independent 10-second hold windows
4. Filter Samples	Automatically exclude high-motion frames and uncalibrated readings before threshold fitting
5. Accept Model	Enforce strict quality gate: both training and validation balanced error must be ≤ 20.0%
6. Save Locally	Persist labeled time-series samples to `.csv` and learned cutoff parameters to `.json`

Table 4.1: Six-Stage Personalized AI Training, Motion Filtering, and Validation Pipeline

4.4 Discussion and Build Progress vs. Plan

Table 4.2 maps the current implementation status of PosturePulse against the 50-mark project evaluation rubric, documenting completed software/firmware artifacts alongside the immediate physical verification steps.

Planned Work & Criterion (Marks)	Current Verified Evidence	Next Verification Milestone
Circuit & Parts (Build Progress: 15M)	Complete BOM, ESP32/Uno pin mapping, and level-shifted wiring guide ready	Verify physical sensor breakouts and 3.3V load continuity
Embedded Firmware (Tech Approach: 10M)	C++ sketch prepared for dual MPU6050s (`0x68`/`0x69`), OLED (`0x3C`), and haptics	Compile, flash to microcontroller, and run bench timing tests
Personalized AI (Data & Model: 10M)	Supervised 1D threshold trainer (5°–40° sweep) and CSV/JSON logger coded	Collect labeled multi-wearer examples during live trials

Planned Work & Criterion (Marks)	Current Verified Evidence	Next Verification Milestone
Software Checks (Functionality: 10M)	Synthetic classifier convergence and serial parser unit checks passed	Verify live USB serial and BLE GATT integration
Teamwork & Design Response (5M)	Added voltage translation checks, dual-sensor curvature math, and corrected wiring guide	Record physical 5-stage wearer demonstration video

Table 4.2: AIBOTS-2026 Review 2 Evaluation Scheme and Project Build Progress vs. Plan



Figure 4.1: ESP32 Bluetooth Low Energy (BLE) Telemetry and Mobile Dashboard Ecosystem

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project designed and documented **PosturePulse**, an interdisciplinary ergonomic posture monitoring and biofeedback system integrating Analog Electronics, Digital Signal Processing, and Machine Learning. By pairing an upper thoracic IMU (`0x68`) with a lower lumbar reference IMU (`0x69`) on a shared 3.3V I²C bus and fusing accelerometer gravity vectors with gyroscope angular rates via a discrete complementary filter, the system accurately tracks sagittal spinal curvature while rejecting high-frequency motion noise and whole-torso forward hip leans. Furthermore, the supervised 1D threshold classifier personalizes the reminder angle (5°–40°) to each individual wearer under a strict ≤20% balanced error gate, while the 10-second continuous timer, 5° hysteresis deadband, and I²C fault latch guarantee safe, chatter-free operation.

5.2 Current Limitations

1. **Physical Multi-Wearer Dataset Collection:** While the C++ firmware, wiring guide, and Python supervised trainer have passed synthetic model and parser verification, systematic accuracy logging across diverse human subjects remains to be recorded during physical bench trials.
2. **Single-Plane Sagittal Tracking:** The current pitch formulation focuses on forward sagittal flexion (“tech neck” and thoracic slouching) and does not yet evaluate coronal-plane lateral leaning (roll asymmetry).

3. **Rigid Breakout Harnessing:** Standard breadboard jumper cables between the upper and lower spine nodes can experience mechanical tug during vigorous torso rotation unless strain-relieved inside a flexible textile strap.

5.3 Future Enhancements

- **Full BLE Mobile App Dashboard:** Complete integration of the ESP32 BLE GATT service with a cross-platform Flutter / React Native mobile dashboard featuring a Daily Posture Score (0–100), shift-level ergonomic heatmaps, and over-the-air customization of alert delay timers (5s, 10s, 30s) and vibration intensity.
- **2D Sagittal + Coronal Classifier:** Expand the supervised learning feature vector to include both relative pitch curvature ($\Delta\theta_{pitch}$) and lateral roll asymmetry ($\Delta\varphi_{roll}$) using a multi-feature Random Forest or TinyML neural classifier running directly in ESP32 flash memory.
- **Flexible Printed Circuit Strap:** Replace discrete jumper wires with a custom flexible PCB spine ribbon housing the ESP32-C3/S3, dual MPU6050 dies, and coin motor inside a breathable neoprene harness.
- **Edge TinyML On-Device Training:** Port the Python supervised threshold sweep directly into ESP32 Non-Volatile Storage (NVS) so that wearers can train, validate, and update their personal posture cutoff angle completely standalone using only the tactile button and OLED display without requiring a laptop USB connection.
- **Longitudinal Ergonomic Analytics:** Integrate automated Pomodoro sitting-break reminders and weekly spinal fatigue trend reports to correlate work-shift posture scores with workstation monitor height and chair lumbar adjustments.

Validation Dimension	Quantitative Evaluation Metric	Target Benchmark for Final Deployment
Angle Estimation Accuracy	Root Mean Square Error (RMSE) vs. optical goniometer	RMSE ≤ 1.5° across 0°–45° flexion range
Hip-Lean Rejection Ratio	False alarm rate during 20 intentional desk reaches	< 5% false positive triggers during 15°–25° hip leans
AI Generalization	Out-of-sample validation balanced error across N ≥ 5 wearers	Balanced error ≤ 12% (strictly within 20% gate)
Timer & Hysteresis Precision	Hold duration before alert onset and reset latency	10.0s ± 0.2s trigger delay; < 250 ms hysteresis silence
Battery Runtime	Continuous active monitoring on 500 mAh 3.7V LiPo cell	≥ 14 hours continuous operation per USB-C charge

Table 5.1: Proposed Evaluation Metrics for Multi-Subject Ergonomic and Hardware Validation

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